

Document 04 — UI / UX Design Brief

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Direction

VelaDial should feel like a small, quiet bedside or door-side instrument rather than a flashy dashboard. The display is round, small, and used in a dark bedroom, so the interface should be minimal and highly legible.

Visual style

- Dark-mode only.
- Pure black background.
- Amber accent color.
- White text for high contrast.
- Large readable typography.
- Simple round-screen geometry.
- Minimal motion.
- No bright full-screen white states.
- Performance-safe for ESPHome LVGL.

Color palette

PURPOSE	COLOR	NOTES
Background	#000000	Helps blend into the black bezel.
Primary active	#FFB300	Warm amber, used for brightness and active glow.
Text	#FFFFFF	Keep high contrast.
On state	#4CAF50	Use sparingly.

PURPOSE	COLOR	NOTES
Off/error state	#F44336	Use sparingly; avoid alarm-like visuals.

Typography

- Use Roboto or a similarly clean sans-serif font.
- Primary values should be large enough to read quickly.
- Avoid long labels; the display is too small for paragraph UI.

Main UI pages

The main door-side UI has exactly 3 primary control pages, plus a hidden device-level theme selector:

1. Power
2. Brightness
3. Presets
4. Theme Selector (hidden, accessed via long-press)

Interaction model

Rotary input

The physical rotation should feel directly connected to the brightness arc. The display should update immediately, even if the Home Assistant command is debounced slightly.

Touch input

Touch should be reserved for major actions:

- Wake screen.
- Toggle power.
- Select preset.
- Move between pages.

Sleep behavior

The display should dim after inactivity. Waking the display should not accidentally change light state unless the user performs a deliberate second action.

Page navigation

- Horizontal swipe navigates between pages.
- Left swipe = next page (Power → Brightness → Presets → Power).
- Right swipe = previous page.
- A 3-dot page indicator appears near the bottom of the round screen.
- The active page's dot is amber; inactive dots are low-contrast white.
- There is **no** 4th Environment page in the first build.

Knob press behavior per page

Knob press is page-specific and applies only when the display is awake:

- **Power page:** knob press toggles the bedroom light group (same as tapping the center).
- **Brightness page:** knob press returns to the Power page.
- **Presets page:** knob press applies the currently highlighted preset.
- **Theme Selector (any page):** long-press (>1.5s) enters the Theme Selector.
- **While asleep:** knob press wakes only.

UI screens

Power

- Large central power symbol or state label.
- Small status text: On / Off / Unavailable.
- Optional amber ring when lights are on.

Brightness

- Arc or circular progress indicator.
- Large percentage value in the center.
- Rotary encoder is the primary input.

Presets

First-build presets are locked to exactly 4:

1. Warm White
2. Soft Amber
3. Neutral White
4. Low Nightlight

Layout:

- 2×2 large touch targets on the 240×240 round screen.
- Each tile sized for thumb-tap; tiles stay inside the inscribed area so corners are not cut off by the round bezel.
- Active preset is highlighted with an amber border (or equivalent amber highlight).

Out of first-build scope:

- Optional RGB accent preset is future / v2 only.
- No dense color picker, color wheel, or per-bulb editor in first build.

Ambient-aware display behavior

TSL2591 controls adaptive backlight behavior on the door-side display.

- Display should be brighter during daylight/bright room conditions.
- Display should be very dim in dark bedroom conditions.
- Backlight transitions must be smooth, not flickery.
- Ambient light affects display/backlight only, not room bulbs directly.
- Night safety is more important than maximum brightness.
- First interaction may wake the display, but night brightness should still be capped.

Day / dim / night UI modes

MODE	TRIGGER	BACKLIGHT	UI BEHAVIOR
Bright	TSL2591 lux > 50	High	Normal contrast, readable from door-side distance.
Dim	TSL2591 lux 5–50	Moderate	Amber/white contrast preserved.
Night	TSL2591 lux < 5	Low	Minimal glow, no large bright fills.

The UI layout should not change dramatically between modes; only brightness/intensity should adapt.

LED ring brightness behavior

This section describes the **door-side WS2812 LED ring** built into the ELECROW rotary display module. It is a separate system from the optional **bedside status LED** described in [docs/03_App_Flow.md](#) and is independently controlled.

- LED ring should stay off when room lights are off.
- When lights are on, LED ring may glow amber at low brightness.
- LED ring brightness may be capped by ambient-light mode.
- No flashing error patterns at night.
- No aggressive animations.
- Optional future: ambient-aware LED ring intensity.
- **Theme-colored:** The LED ring color matches the currently active UI theme (e.g., blue for Minimal Thermostat, gold for Eclipse Corona).

Bedside sensor UX

The bedside controller has no screen, so its UX is entirely physical sensors plus an optional silent LED. v1 and v2 are split to match [docs/03_App_Flow.md](#).

v1 (first build)

- **APDS-9960 standalone left/right gestures.** APDS-9960 owns directional gestures without being armed or gated by VL53L4CD.
- **VL53L4CD standalone hand-hold nightlight** (if VL53L4CD ESPHome support is verified). Nightlight hold must feel deliberate: stable hand at 5–10 cm for about 1.5 s.
- Quick pass-by motion must not trigger nightlight.
- Cooldowns prevent repeated actions.
- No audio, no voice, no beeps.

v2 / future (not required for first firmware)

- VL53L4CD arms, wakes, or refines APDS-9960 gesture detection (sensor fusion).
- Sensor fusion is **not** required for the first firmware build. It should be added only after both v1 standalone paths are verified to work reliably in the actual bedside mounting.

Environmental data UX

SHT45 temperature/humidity is secondary.

- Do not place temp/humidity on the main Power / Brightness / Presets pages as a required element.
- There is **no** required 4th Environment page in the first build.
- A diagnostics view, idle mini-status, or long-press info overlay are **future / v2 possibilities only, not first build**.
- Environmental data is exposed to Home Assistant and can be shown in the Home Assistant dashboard.
- Environmental data must not compete with lighting controls.
- No HVAC / comfort automation UX is required for the first build.

Wake and accidental-trigger UX

The following wake behavior is locked for the first build (matches [docs/03_App_Flow.md](#)):

- **Touch while asleep:** wake only.
- **Knob rotation while asleep:** wake only.
- **Knob press while asleep:** wake only.
- A second deliberate action while the display is awake is what toggles or sends a command.
- Avoid accidental room-light changes from bumps.
- Bedroom safety is more important than shortcut speed.
- Sensor readings must not cause sudden full-brightness room light changes.

Offline / unavailable visual state

When Home Assistant, the Raspberry Pi, or the local LAN is unreachable, the door-side display should clearly indicate the state without aggressive UX. This matches the connectivity behavior in [docs/03_App_Flow.md](#).

- Use **"Unavailable"** as the preferred UI wording on the door-side display.
- Show a subtle, persistent badge or small label rather than a full-screen overlay.
- Do **not** pretend a command succeeded without state confirmation from Home Assistant.
- Dim or desaturate affected controls if useful for clarity (for example, gray the brightness percentage, mute the power-state icon).
- **No** flashing.
- **No** popup modal.

- **No** audio.
- **No** full-screen aggressive error.
- The offline / unavailable state belongs mainly on the door-side display, not on a constantly lit bedside LED.

Fallback UX

- If TSL2591 fails, use fixed dim/sleep behavior.
- If SHT45 fails, hide environmental data.
- If VL53L4CD fails, fall back to APDS-9960 gesture-only behavior if possible.
- If APDS-9960 fails, ToF nightlight hold may still work if VL53L4CD works.
- Fail quietly and recover automatically.
- Show errors only in diagnostics, not with flashing LEDs.

LVGL feasibility guidance

- Prefer simple arcs, labels, circles, and large buttons.
- Avoid tiny widgets and dense dashboard layouts.
- Avoid complex 3D, heavy gradients, and excessive animation in final firmware.
- AI-generated visual concepts can inspire the design, but final UI must be practical for ESPHome LVGL on a 240×240 round display.
- The production firmware contains 20 selectable UI themes. Each theme must individually remain within LVGL performance constraints.
- Theme switching occurs at runtime without reflashing; all 20 themes share the same LVGL widget pool.

Available UI Themes (20 total)

The production firmware includes the following selectable themes, each with a unique visual metaphor and LED ring color:

#	THEME NAME	LED RING COLOR
0	Minimal Thermostat	Blue <code>#4FC3F7</code>
1	SmartKnob Arc	Amber <code>#FFB000</code>
2	Large Center Power	Green <code>#4CAF50</code>

#	THEME NAME	LED RING COLOR
3	Single-Page Simple	Warm #FFB74D
4	Preset Ring	Amber #FFB000
5	Night Mode Ultra-Min	Deep Red #FF4400
6	Text-First Utility	Amber #FFB000
7	Apple Watch Complications	Blue #4FC3F7
8	LED-Ring Status	Amber #FFB000
9	Three-Screen Carousel	Amber #FFB000
10	Brightness-First	Amber #FFB000
11	Door Switch Replace	Green #4CAF50
12	Lunar Phase	Moon White #E8E0D0
13	Sundial Shadow	Gold #FFB000
14	Tree Ring Growth	Wood #FFD080
15	Topographic Contour	Amber #FFB000
16	Iris Aperture	Amber #FFB000
17	Radar Sweep	Amber #FFB000
18	Vinyl DJ Crossfader	Amber #FFB000
19	Eclipse Corona	Gold #FFB000

Design references

- Smart thermostat rotary control clarity.
- Home Assistant dark theme simplicity.
- SmartKnob-style relationship between physical dial and visual feedback.

These are references, not skins to copy exactly.